Network Troubleshooting, Management, Monitoring, and Optimization

Task 1. Use the Windows command prompt to run the command ipconfig /all on your computer and take a screenshot of the output. Identify and write down your IPv4 address, subnet mask, default gateway, and DNS servers as shown in the result.

Answer:

`ipconfig /all`
In the adapter you are using (Wi-Fi or Ethernet), find these lines:

Information	Line to look for / my value
IPv4 Address	10.123.18.212 (Wi-Fi adapter)
Subnet Mask	255.255.255.0
Default Gateway	10.123.18.168 (IPv6 gateway: fe80::b098:7dff:fe8c:27d4)
DNS Servers	10.123.18.168 and 2401:4900:78f5:a8f9::70



```
C:\Users\Admin>
C:\Users\Admin>ipconfig/all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : DESKTOP-LR1PVJE
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : 6C-94-66-47-0D-1B
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

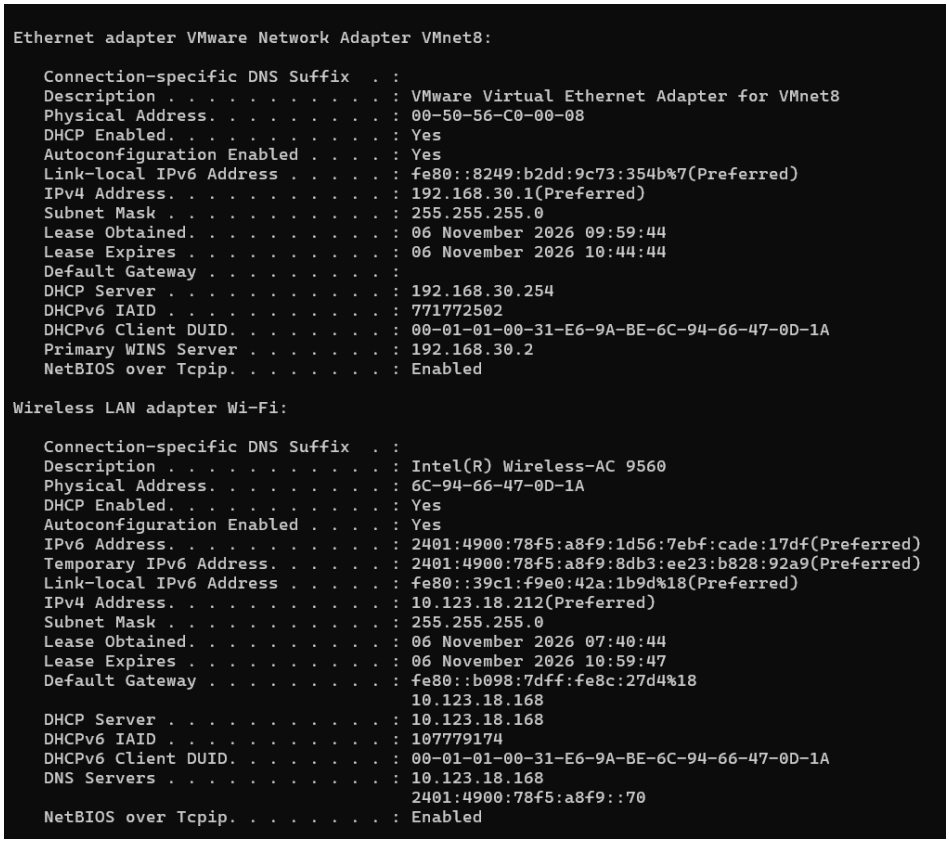
Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
   Physical Address. . . . . . . . . : 6E-94-66-47-0D-1A
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes

Ethernet adapter VMware Network Adapter VMnet1:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VMware Virtual Ethernet Adapter for VMnet1
   Physical Address. . . . . . . . . : 00-50-56-C0-00-01
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::3b27:d168:dbde:9332%16(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.133.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : 06 November 2026 09:59:52
   Lease Expires . . . . . . . . . . : 06 November 2026 10:44:44
   Default Gateway . . . . . . . . . :
   DHCP Server . . . . . . . . . . . : 192.168.133.254
   DHCPv6 IAID . . . . . . . . . . . : 738218070
   DHCPv6 Client DUID. . . . . . . . : 00-01-01-00-31-E6-9A-BE-6C-94-66-47-0D-1A
   NetBIOS over Tcpip. . . . . . . . : Enabled
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Screenshot 1: ipconfig /all (host name and VMware adapters)



Screenshot 2: ipconfig /all (Wi-Fi adapter with IPv4, subnet mask, gateway and DNS servers)



Screenshot 3: ipconfig /all (Bluetooth adapter)

The values above are from the "Wireless LAN adapter Wi-Fi" section (Intel Wireless-AC 9560), which is the adapter connected to the internet. The VMware adapters (192.168.133.1 and 192.168.30.1) are virtual adapters and have no default gateway.

Meaning: IPv4 address is the address of your computer in the network; the subnet mask tells which part of the address is the network; the default gateway is the router that connects you to other networks and the internet; the DNS server converts website names to IP addresses.

Task 2. Simulate a network connectivity issue by disconnecting your Wi-Fi or Ethernet cable, then use the ping command to check connectivity to www.google.com. Reconnect and repeat the ping. Write a short note on the difference in results and what each status means.

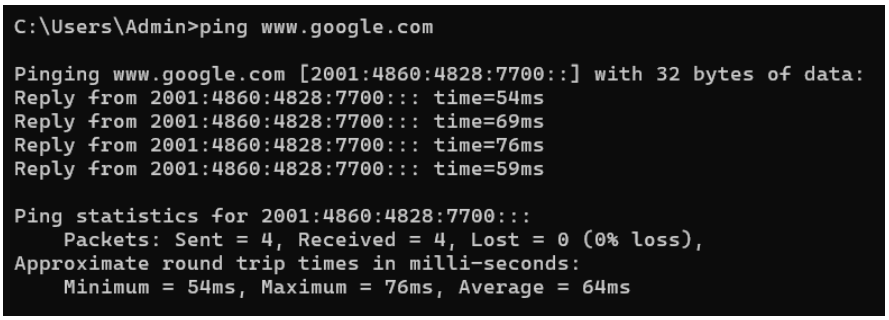
Answer:

1. Turn off Wi-Fi (or unplug the Ethernet cable) and run: ping www.google.com
2. Note the message that appears.
3. Reconnect Wi-Fi / the cable, wait a few seconds, and run the same command again.
4. Compare the two results.

	While disconnected	After reconnecting
Typical result	An error such as "Ping request could not find host www.google.com", "General failure", "Destination host unreachable" or "Request timed out". No replies.	Replies such as "Reply from <IP>: bytes=32 time=20ms TTL=117", and at the end "0% loss".
My result	Ping request could not find host www.google.com. Please check the name and try again.	Reply from 2001:4860:4828:7700:: with times of 54, 69, 76 and 59 ms. Packets: Sent = 4, Received = 4, Lost = 0 (0% loss). Min = 54 ms, Max = 76 ms, Average = 64 ms.



Screenshot 1: ping www.google.com while Wi-Fi was disconnected



Screenshot 2: ping www.google.com after reconnecting to Wi-Fi

Short note (difference and meaning):

In my test, with Wi-Fi off the ping gave the error "Ping request could not find host www.google.com. Please check the name and try again." This means the computer could not turn the name into an IP address, because there was no network connection to reach a DNS server. After reconnecting, the same ping worked: all 4 packets got a reply, there was 0% packet loss and the average time was 64 ms, so the internet connection and DNS were working.

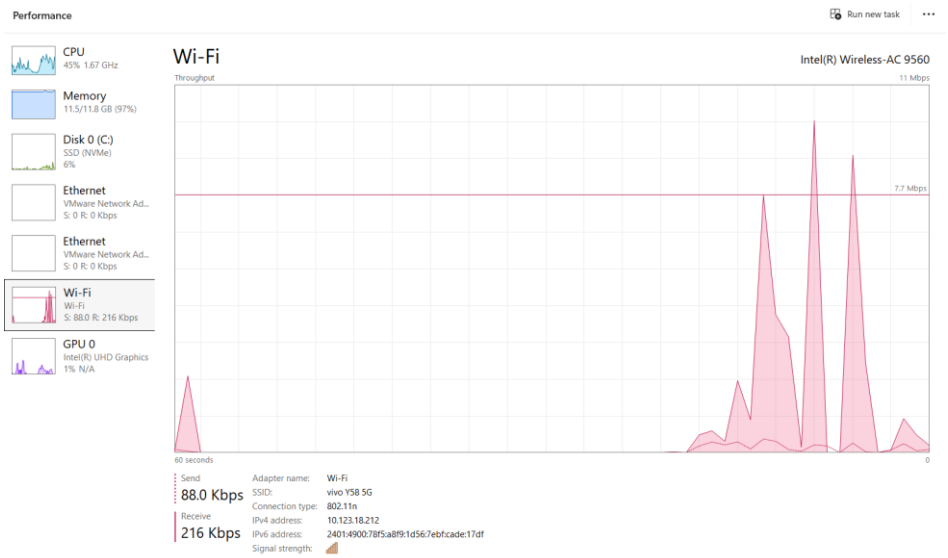
In general: when the connection is disconnected, the ping fails because the computer has no network path to send the packets, or it cannot find the IP address of the website because DNS cannot be reached. The error means there is no connectivity (a local problem). After reconnecting, the ping gets replies with a time in milliseconds and 0% packet loss, which means the computer can reach Google, DNS is working and the internet connection is healthy. "Request timed out" means the packets were sent but no reply came back, and "Destination host unreachable" means there is no route to reach the destination.

Task 3. Open Task Manager or Resource Monitor on your system and monitor network usage for 5 minutes while streaming a YouTube video. Record the peak network usage value and note which process used the most bandwidth during this time.

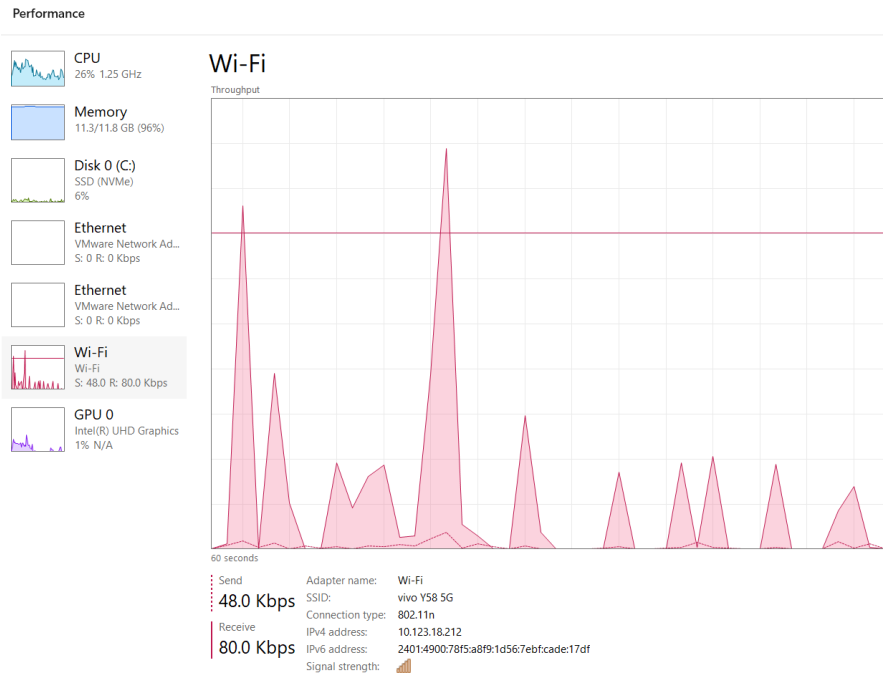
Answer:

1. Open Task Manager (Ctrl + Shift + Esc) and go to the Performance tab, then select Wi-Fi or Ethernet. You will see the network usage graph and the current Send / Receive speed.

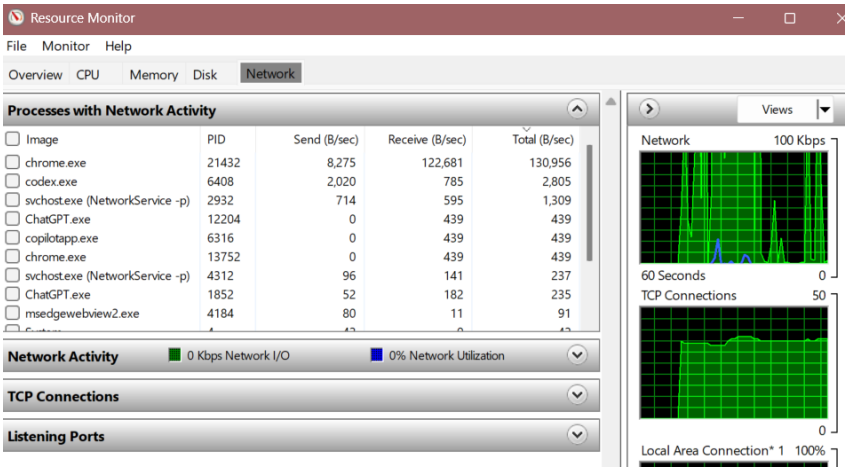
2. Also open Resource Monitor (type resmon in the Start menu) and go to the Network tab, where "Processes with Network Activity" shows which process uses the network.
3. Start a YouTube video in the browser and let it play for 5 minutes. Keep watching the graph and note the highest value (peak).
4. In the Processes tab of Task Manager, click on the Network column to sort and see which process uses the most bandwidth.



Screenshot 1: Task Manager > Performance > Wi-Fi graph while streaming YouTube (graph scale up to 11 Mbps)



Screenshot 2: Task Manager Wi-Fi graph during the video



Screenshot 3: Resource Monitor > Network tab, sorted by Total (B/sec) while YouTube was playing

Peak network usage: about 10 Mbps. In screenshot 1 the highest spike on the Wi-Fi graph reaches almost the top of the graph, and the top of the graph is 11 Mbps. The graph also shows a marker line at 7.7 Mbps. The usage was not constant: it came in short bursts, because YouTube loads the video in parts (buffering). Between the bursts the Wi-Fi usage was low, for example Send 48 Kbps and Receive 80 Kbps in screenshot 2.

Processes with network activity (from Resource Monitor, sorted by Total B/sec):

Process	PID	Send (B/sec)	Receive (B/sec)	Total (B/sec)
chrome.exe	21432	8,275	122,681	130,956
codex.exe	6408	2,020	785	2,805
svchost.exe (NetworkService -p)	2932	714	595	1,309
ChatGPT.exe	12204	0	439	439
copilotapp.exe	6316	0	439	439
chrome.exe	13752	0	439	439
svchost.exe (NetworkService -p)	4312	96	141	237
ChatGPT.exe	1852	52	182	235
msedgewebview2.exe	4184	80	11	91

Process that used the most bandwidth: chrome.exe (Google Chrome, PID 21432) used the most bandwidth. Its total was 130,956 B/sec (Send 8,275 B/sec and Receive 122,681 B/sec), which is about 1 Mbps. This is far higher than the next process, codex.exe (2,805 B/sec). It is expected, because the YouTube video was playing in Chrome, and the video data is downloaded (received) by the browser. That is why the Receive value is much larger than the Send value.

Note: Resource Monitor shows the speed in bytes per second at that moment, and it changes every second, so the numbers will be different each time. The Task Manager Wi-Fi graph is better for seeing the peak, and Resource Monitor is better for seeing which process is using the network.

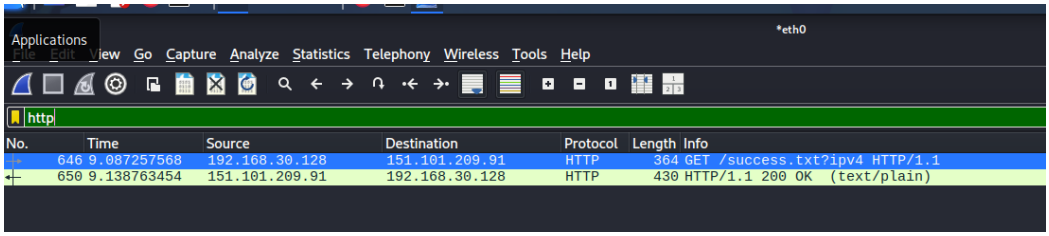
Task 4. Download and install Wireshark (or use an online packet capture demo if installation is not possible). Capture at least 30 seconds of network traffic while

browsing Flipkart or Instagram Web, then filter the packets to show only HTTP or HTTPS traffic. List 3 types of information you can see in these filtered packets.

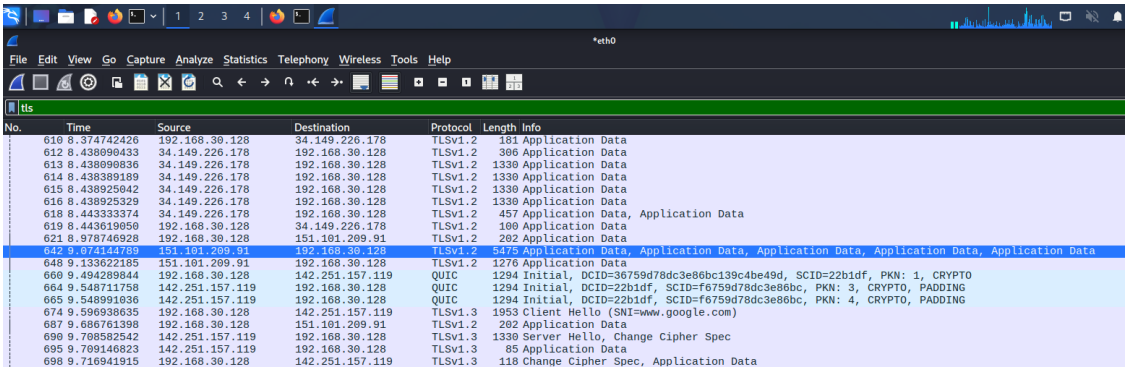
Hint: *Use the filter bar in Wireshark and try http or https as the filter keyword.*

Answer:

1. Install Wireshark (wireshark.org) and open it. Double-click your active network adapter to start the capture.
2. Browse Flipkart or Instagram Web for at least 30 seconds, then click the Stop button.
3. In the display filter bar, type http and press Enter to see plain HTTP traffic. For HTTPS, type tls (or tcp.port == 443), because HTTPS runs over TLS and Wireshark shows it as TLS.
4. Click on a few filtered packets and look at the details panel.



Screenshot 1: Wireshark with the filter "http" (plain HTTP packets only)



Screenshot 2: Wireshark with the filter "tls" (encrypted HTTPS traffic)

I used the Wireshark capture on interface eth0 (the packets were captured while browsing). With the filter "http" only 2 packets are shown (a request and its reply), and with the filter "tls" many HTTPS packets are shown.

3 types of information visible in the filtered packets:

1. Source and destination IP addresses: for example 192.168.30.128 (my computer) talking to 151.101.209.91 and 34.149.226.178, which shows which computer is talking to which server. Port numbers are also visible (HTTPS uses port 443).
2. Protocol and handshake details: in the tls filter I can see the TLS version (TLSv1.2 and TLSv1.3) and handshake steps such as "Client Hello (SNI=www.google.com)", "Server Hello, Change Cipher Spec". The SNI shows the name of the website I am connecting to, even though the data is encrypted.
3. Request and response details of plain HTTP: in the http filter I can see the request "GET /success.txt?ipv4 HTTP/1.1" and the reply "HTTP/1.1 200 OK (text/plain)", which shows the method, the page requested, the status code and the content type. Packet length and time are also shown for each packet (for example 364 and 430 bytes).

Note: The actual content of HTTPS packets (like passwords or pages) is shown only as "Application Data" because it is encrypted, so Wireshark cannot read it. Most sites like Flipkart

and Instagram use HTTPS, so there is very little plain HTTP. The one HTTP request seen here looks like a small connectivity check made by the browser.

Task 5. Imagine your friend is complaining about slow internet while gaming online. Use ChatGPT or any AI tool to generate a troubleshooting checklist (at least 5 points) for optimizing home network speed and reliability. Paste the AI's response and highlight one suggestion you would try first.

Answer:

Troubleshooting checklist (the highlighted point is the one I would try first):

No.	Suggestion	Details
1	Use a wired Ethernet connection (try this first)	Connect the PC or console to the router with an Ethernet cable instead of Wi-Fi. A cable gives lower ping, less lag and a more stable connection than Wi-Fi.
2	Restart the router and modem	Switch them off for 30 seconds and turn them on again to clear errors and refresh the connection.
3	Close background downloads and streaming	Stop other devices and apps that use bandwidth (video streaming, game updates, cloud backup) while gaming.
4	Improve the Wi-Fi signal	If Wi-Fi must be used, move closer to the router, use the 5 GHz band, and avoid walls and electrical devices in between.
5	Check speed and ping	Run a speed test and ping the game server or 8.8.8.8. High ping or packet loss shows a network problem.
6	Update the router firmware and drivers	Install the latest firmware for the router and the latest network driver on the PC.
7	Use QoS or change the DNS	Turn on QoS (Quality of Service) in the router to give gaming traffic priority, and try a faster DNS like 8.8.8.8 or 1.1.1.1.
8	Contact the ISP	If the speed is still slow after all these steps, call the ISP to check the line or the plan.

Why I would try point 1 first: It is quick and free, and Wi-Fi is the most common reason for lag and ping spikes in online games.